

Harik Sodhi

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EDUCATION

University of Oxford (Christ Church) | *MEng Engineering Science* Oct 2025 – Jun 2029

AWARDS & HONOURS

- **Silver Medal, International Physics Olympiad** (2025) – 49th/420 competitors from 89 countries; highest-ranked competitor from Western Europe.
- **United Kingdom International Physics Olympiad Team Member** (2025) – Selected after Gold awards in BPhO Round 2 and BAAO Round 2.
- **Winner, Berkeley Math Tournament Online Power Round** (2024) – First place in a proof-based competition on graph theory, cryptography, and computational complexity.
- **Ritangle National Champion** (2023) – Team Captain; 1st/1,200 teams in a national combinatorial optimisation challenge, developing and proving an optimal solution to a highly constrained scheduling problem with hundreds of interacting variables.

EXPERIENCE

Oxford Alpha Fund Oxford
Quant Analyst / Director Jan 2026 – Present

- Selected through a competitive quantitative research and presentation process as a first-year undergraduate.
- Researched volatility risk premia in short-dated Bitcoin options using historical Deribit/Coinbase data; developed and backtested systematic trading strategies achieving a Sharpe ratio of 1.18 and a 70% win rate. [Code]
- Built automated tooling to generate sponsor-ready CV books from member CVs, streamlining recruitment preparation.
- Coordinated with sponsoring firms and industry partners to support recruitment events and strengthen fund relationships.

Optiver Trading Academy Oxford
Participant Oct 2025 – Nov 2025

- Selected for Optiver's competitive Trading Academy programme as a first-year undergraduate.
- Traded ETFs and futures on Optiver's proprietary simulation platform, applying market-making, delta-hedging, and risk-management principles under realistic market conditions.

LEADERSHIP & ACTIVITIES

VEX Robotics & REC Foundation London / Remote
Global Student Advisory Board Member & Team Captain May 2021 – Mar 2024

- Selected as one of 12 members globally to directly advise the CEO and executive leadership team on the strategic direction and accessibility of the VEX Robotics programme.
- Served as Team Captain, Lead Programmer, and Lead Designer; managed team operations and spearheaded end-to-end development of competitive robots.
- Developed autonomous routines and driver-assist systems in C++ using PID control and designed competition mechanisms in CAD.

PROJECTS

Market Making Simulator

- Developed a multi-agent market simulation modelling market makers, noise traders, and informed traders.
- Analysed the effects of participant composition on spread formation, adverse selection, and trading profitability.
- Implemented strategy evaluation and performance analysis tools for comparing trading behaviour. [GitHub]

Rocket Lander

- Designed and developed a MATLAB simulation of autonomous rocket landing dynamics from first principles, incorporating variable-mass propulsion, atmospheric drag, and gravity.
- Implemented Runge-Kutta numerical integration, a scratch-built PID control system, and constrained optimisation techniques to achieve stable autonomous touchdowns. [GitHub]

SKILLS

- **Programming:** Python, C++, MATLAB, LaTeX
- **Quantitative:** Statistical Modelling, Backtesting, Optimisation, Stochastic Simulation, Volatility Analysis, Numerical Methods
- **Libraries:** NumPy, Pandas, SciPy, PyTorch, Scikit-learn, Matplotlib
- **Tools:** Linux, Git, GitHub, SolidWorks